A VARIANT OF THE CRITERION FOR PROPERTIES OF COHERENT SHEAVES, TESTING ON SHEAVES WITH PRESCRIBED SCHEME THEORETIC SUPPORT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The initial criterion reducing a property of coherent sheaves to two conditions on the ideals of reduced irreducible subspaces. Same chapter spaces-cohomology.tex, label lemma-property-initial (invoked at proof line 2797). The proof only has to verify conditions (1) and (2) of that criterion; condition (1) is immediate and condition (2) is what the minimal counterexample argument establishes.
- Preparation of a coherent module on a reduced irreducible subspace by a pushed forward ideal power. Same chapter spaces-cohomology.tex, label lemma-prepare-filter-irreducible (invoked at proof line 2817). It provides the injective map $\varphi : i_*\mathcal{I}^{\oplus r} \to \mathcal{G}$ used to compare the two sheaves.
- The filtration of a coherent sheaf on a Noetherian space with subquotients pushed forward from ideals on irreducible supports. Same chapter spaces-cohomology.tex, label lemma-coherent-filter (invoked at proof line 2822). The filtration of $\Coker(\varphi)$ lets minimality of Z together with condition (1) give property $\mathcal{P}$ for that cokernel.

Source reference. Original lemma 07UR: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2521–2521. Navigation pointer for \ref{lemma-coherent-filter} used inside the counted span (line 2822); the argument that consumes it is retained in the span.

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Source reference. Original lemma 07UQ: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2478–2478. Navigation pointer for \ref{lemma-prepare-filter-irreducible} used inside the counted span (line 2817); the argument that consumes it is retained in the span.

Source reference. Original lemma 07US: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2616–2616. Navigation pointer for \ref{lemma-property-initial} used inside the counted span (line 2797); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S. Let $\mathcal{P}$ be a property of coherent sheaves on X. Assume*

1. *For any short exact sequence of coherent sheaves on X if two out of three have property $\mathcal{P}$ so does the third.*
2. *If $\mathcal{P}$ holds for $\mathcal{F}^{\oplus r}$ for some $r \geq 1$, then it holds for $\mathcal{F}$.*
3. *For every reduced closed subspace $i : Z \to X$ with $|Z|$ irreducible there exists a coherent sheaf $\mathcal{G}$ on X whose scheme theoretic support is Z such that $\mathcal{P}$ holds for $\mathcal{G}$.*

Then property $\mathcal{P}$ holds for every coherent sheaf on X.

Proof. We will show that conditions (1) and (2) of Lemma 07US hold. This is clear for condition (1). To show that (2) holds, let

$$\mathcal{T} = \left\{ \begin{array}{l} i : Z \to X \text{ reduced closed subspace with } |Z| \text{ irreducible such} \\ \text{ that } i_*\mathcal{I} \text{ does not have } \mathcal{P} \text{ for some quasi-coherent } \mathcal{I} \subset \mathcal{O}_Z \end{array} \right\}$$

If $\mathcal{T}$ is nonempty, then since X is Noetherian, we can find an $i : Z \to X$ which is minimal in $\mathcal{T}$. We will show that this leads to a contradiction.

Let $\mathcal{G}$ be the sheaf whose scheme theoretic support is Z whose existence is assumed in assumption (3). Let $\varphi : i_*\mathcal{I}^{\oplus r} \to \mathcal{G}$ be as in Lemma 07UQ. Let

$$0 = \mathcal{F}_0 \subset \mathcal{F}_1 \subset \ldots \subset \mathcal{F}_m = \text{Coker}(\varphi)$$

be a filtration as in Lemma 07UR. By minimality of Z and assumption (1) we see that $\text{Coker}(\varphi)$ has property $\mathcal{P}$. As φ is injective we conclude using assumption (1) once more that $i_*\mathcal{I}^{\oplus r}$ has property $\mathcal{P}$. Using assumption (2) we conclude that $i_*\mathcal{I}$ has property $\mathcal{P}$.

Finally, if $\mathcal{J} \subset \mathcal{O}_Z$ is a second quasi-coherent sheaf of ideals, set $\mathcal{K} = \mathcal{I} \cap \mathcal{J}$ and consider the short exact sequences

$$0 \to \mathcal{K} \to \mathcal{I} \to \mathcal{I}/\mathcal{K} \to 0 \quad \text{and} \quad 0 \to \mathcal{K} \to \mathcal{J} \to \mathcal{J}/\mathcal{K} \to 0$$

Arguing as above, using the minimality of Z, we see that $i_*\mathcal{I}/\mathcal{K}$ and $i_*\mathcal{J}/\mathcal{K}$ satisfy $\mathcal{P}$. Hence by assumption (1) we conclude that $i_*\mathcal{K}$ and then $i_*\mathcal{J}$ satisfy $\mathcal{P}$. In other words, Z is not an element of $\mathcal{T}$ which is the desired contradiction. □